YOUSEF ELSHAZLY

Data Analyst & ML Engineer

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Profile

Data-focused graduate with hands-on experience building pipelines, working with messy real-world data, and applying ML to content and search problems. Currently working on content ranking and search signal analysis at FlyRank. Comfortable going from raw data to a working model — focused on output that is useful and honest, not just functional.

Experience

FlyRank July 2026 – Present
ML Intern – Applied Search Intelligence

- Analysed a 30,000-page search dataset across 32 clients to build a content refresh prioritisation model using Python and pandas.
- Identified and documented label leakage risk in training data; reformulated target variable to use strictly prior-period features.
- Benchmarked model against simple rule baseline — model precision@50: 0.740 vs 0.240 for the rule baseline.

Raka-Design October 2024 – April 2025
Data Analyst & Engineer

- Developed data pipelines using Python (pandas, NumPy) to automate workflows and manage data efficiently.
- Integrated various APIs using Python, JSON, and JavaScript to fetch, process, and sync data across platforms.
- Collaborated with design and development teams to support data-driven decisions.

Schneider Electric August 2024 – September 2024
Business Development Intern

- Analysed pricing models, calculating changes and revenue impact.
- Designed a brand page on a leading e-commerce platform, improving online presence.
- Acted as Data Steward, ensuring data quality through governance policies and supporting sales teams by managing KPIs using Salesforce.

Roadtech February 2024 – September 2024
Account Manager

- Integrated distributed finances into a centralised database with data consistency.
- Conducted financial audits for accuracy and compliance.
- Implemented data validation processes and ensured data security.

Emmkan July 2021 – June 2022
Intern

- Sales Team: Implemented sales tactics and understood target audiences.
- Workshop Facilitator: Educated high school students on social-emotional skills.
- Contributor to SEL Game: Helped develop a relationship-skills game promoting emotional intelligence.

Projects

Music Recommendation System

- Applied K-Means clustering and PrefixSpan sequential pattern mining to 2,000 Spotify tracks using 18 audio features (energy, valence, tempo, danceability) to group songs by mood and listening pattern.
- Recommended tracks based on cluster membership and sequential listening behaviour; handled multi-genre encoding as a known limitation.
- Evaluated via user study (n=15); 93% of users preferred model recommendations over random selection.

Football Ticket Database System

- Created logical models for database design and implemented them into a SQL database system.
- Utilised Common Table Expressions (CTEs) to extract and aggregate information.
- Developed a user-friendly GUI to visualise and interact with the database.

Skills

Programming: Python (pandas, NumPy, scikit-learn), SQL, Java, JavaScript, HTML
Data & ML: Machine Learning, K-Means Clustering, Data Pipelines, ETL, Data Modelling, Data Analysis, Data Science
Tools & Platforms: Git / Version Control, REST APIs, JSON, MS Project
Databases: Microsoft SQL Server / MySQL, Database Design
Soft Skills: Research, Presentation, Teamwork, Cross-functional Collaboration, Sales

Education

German University in Cairo 2020 – 2024
Bachelor of Science in Business Informatics

- GPA: 3.4 (German scale, / 1.84 on a 0.7 scale).
- Strong foundation in data engineering, software development, databases, and applied machine learning.

Egyptian British International School 2018 – 2020
IGCSE

- Grade: 93.75%